Roadmap. §57 (pp. 1–5): photon path integral. Gauge modes are dropped from $\mathcal{D}A$ (Lorenz/Landau gauge), $P^{\mu\nu} \to g^{\mu\nu}$ on conserved currents (Feynman), R_ξ stated (Srednicki §62). Pp. 6–10 (§22, Noether; skipped in class, optional reading) derive the conserved current that LecNote17 p. 6 states and LecNote21 p. 3 uses. §58 (pp. 11–12): gauging $U(1)$. Next, LecNote23: QED rules (p. 1), R_ξ (pp. 5–6). Supplements: FermionSigns, PhotonGauges.

lec 22.

We heard in the last lecture that photon propagator

$$\Delta(x-y) = \int \frac{d^4k}{(2\pi)^4} e^{ik(x-y)} \Delta^{\mu\nu}(k)$$

$$\Delta^{\mu\nu}(k) = -\frac{1}{k^2} \delta^{\mu 0}\delta^{\nu 0} + \frac{1}{k^2 - i\varepsilon} \sum_{\lambda=\pm} \varepsilon_\lambda^{\mu *}(\vec{k})\, \varepsilon_\lambda^{\nu}(\vec{k})$$

We also heard

$$\sum_{\lambda=\pm} \varepsilon_\lambda^{i *}(\vec{k})\, \varepsilon_\lambda^{j}(\vec{k}) = \delta_{ij} - \frac{k_i k_j}{\vec{k}^2}$$

(recall $\varepsilon_\pm = \frac{1}{\sqrt{2}}(\mp 1, \pm i, 0)$

$\vec{k} = (0, 0, k)$)

Extending to μ, ν ($i \to \mu$, $j \to \nu$)

$$\Delta^{\mu\nu}(k) = \frac{\delta^{\mu\nu}}{k^2 - i\varepsilon}$$ [Feynman Gauge]

~~$\Delta^{\mu\nu}(k) = \frac{\delta^{\mu\nu} - \frac{k^\mu k^\nu}{k^2}}{k^2 - i\varepsilon}$~~ [Coulomb Gauge]

but this is somewhat alarming already as we know photons are massless. [1]

[1] What is alarming: $g^{\mu\nu} = \sum_\pm \varepsilon_\pm^{\mu *} \varepsilon_\pm^{\nu} - n^\mu n^\nu + l^\mu l^\nu$, with $n = (1,0,0,0)$, $l = (0, \mathbf{k}/|\mathbf{k}|)$: four polarizations, the time-like one with the wrong sign, for a photon with two. Between conserved currents the extra pair gives $-|\tilde{J}^0|^2/\mathbf{k}^2$, the Coulomb term above, with no pole at $k^2 = 0$. Relabelling $i \to \mu$ is not the argument; pp. 2–5 are.

One also naturally asks, why these different choices yield a same result?

Let's Path Integral.

$$Z_0[J] = \int \mathcal{D}A\, e^{iS_0}$$

$$S_0 = \int d^4x \left[-\frac{1}{4} F^{\mu\nu} F_{\mu\nu} + J^\mu A_\mu \right].$$

Into momentum space, we have

$$S_0 = \frac{1}{2} \int \frac{d^4k}{(2\pi)^4} \left[-\tilde{A}_\mu(k) (k^2 g^{\mu\nu} - k^\mu k^\nu) \tilde{A}_\nu(-k) + \tilde{J}^\mu(k) \tilde{A}_\mu(-k) + \tilde{J}^\mu(-k) \tilde{A}_\mu(k) \right]$$

(Recall that $\tilde{A}_\mu(k) = \int \frac{d^4x}{(2\pi)^4} e^{-ikx} A(x)$

$\tilde{J}_\mu(k) = \int \frac{d^4x}{(2\pi)^4} e^{-ikx} J(x)$ [1]

$d^4x\, d^4k\, d^4k'\, e^{-i(k+k')x}$ set $k' = -k$.)

Then we can shift the integration variable $\mathcal{D}A$ by $\tilde{J}$ to make it Gaussian and leave with the source terms. but with one problem that the inverse of $k^2 g^{\mu\nu} - k^\mu k^\nu$ doesn't exist.

Separately $\int dx\, e^{i(Ax^2 + Bx)}$

$= \int d(x + \frac{B}{2A} - \frac{B}{2})^2 e^{-i(x - A^{-1}B/2)^2 A + A^{-1}B/4}$.

1 Should read $\tilde{A}_\mu(k) = \int d^4x\, e^{-ikx} A_\mu(x)$, i.e. $A_\mu(x) = \int \frac{d^4k}{(2\pi)^4} e^{ikx} \tilde{A}_\mu(k)$, likewise $\tilde{J}$: the $(2\pi)^4$ goes with d^4k. Under d^4x it puts $(2\pi)^8$ into S_0 above. The sign of the exponent is immaterial here.

Let's write $k^2 g^{\mu\nu} - k^\mu k^\nu = k^2 P^{\mu\nu}(k)$.

$$P^{\mu\nu}(k) = \frac{k^2 g^{\mu\nu} - k^\mu k^\nu}{k^2} = g^{\mu\nu} - \frac{k^\mu k^\nu}{k^2}$$

this is again a projection matrix that is important.

$$P^{\mu\nu}(k) P_\nu{}^\lambda(k) = P^{\mu\lambda}(k)$$

The only allowed eigenvalues are zero and one.

as

~~$k_\mu P^{\mu\nu}(k) k_\nu = 0$~~

$$P^{\mu\nu}(k) k_\nu = k^\mu - k^\mu = 0.$$

So a 4-vector proportional to k^μ will be an eigenvector with eigenvalue 0.

Then $g_{\mu\nu} P^{\mu\nu}(k) = 4 - \frac{k^2}{k^2} = 3$

the "Trace" as $\sum_\mu P^\mu{}_\mu$

So the other 3 eigenvalues are 1, 1, 1.

To make the matrices invertible, we just need to focus on those with non-zero eigenvalues. the

So in $\mathcal{D}A^2$, ~~with~~ we only integrate over ~~sector~~ states

~~with~~ three states ~~proper~~ orthogonal to k^μ direction.[1]

[1] Why drop the k^μ direction? $\tilde{A}_\mu \to \tilde{A}_\mu + k_\mu f(k)$ is a gauge transformation. It leaves F unchanged and shifts the source term by $f\, k \cdot \tilde{J} = 0$ (conserved current). So S_0 is flat along k: that integral is an infinite, J-independent constant, removed by $Z_0(0) = 1$. Keeping $k \cdot \tilde{A} = 0$ picks one A per gauge orbit: Lorenz gauge.

So here we definined the measure $\mathcal{D}A$ more properly.
Due to such orthogonality, we have

~~$k^\mu \tilde{A}_\mu(k) = 0$~~. $k^\mu \tilde{A}_\mu(k) = 0$.

this is equivalent to the Lorentz Gauge.
For these states, theirm proportional to k^μ do not contribute.
Further, with the shifting of momentum, $k^\mu J_\mu$ are not contributing to the path integral neither, as the current is conserved, $\partial_\mu J^\mu = 0$.

($k_\mu \tilde{J}^\mu(k) = 0$).

And also the matrix $P^{\mu\nu} = g^{\mu\nu} - \frac{k^\mu k^\nu}{k^2}$ is the projection operator to the subspace orthogonal to k^μ within the subspace, $P^{\mu\nu}$ is just the identity matrix.
~~then~~ Hence the inverse is simply $P^{\mu\nu}/k^2$

of $k^2 P^{\mu\nu}$.

Now we have

$$Z_0[J] = \exp\left[\frac{i}{2}\int \frac{d^4k}{(2\pi)^4} \tilde{J}_\mu(k) \frac{P^{\mu\nu}}{k^2 - i\varepsilon} \tilde{J}_\nu(-k)\right]$$

$$= \exp\left[\frac{i}{2}\int d^4x\, d^4y\, J_\mu(k) \frac{P^{\mu\nu}}{k^2 - i\varepsilon} \Delta^{\mu\nu}(x-y) J_\nu(-k)\right]$$ [1]

where

$$\Delta^{\mu\nu}(x-y) = \int \frac{d^4k}{(2\pi)^4} e^{-ik(x-y)} \frac{P^{\mu\nu}}{k^2 - i\varepsilon}$$

1 Should read $\exp\left[\frac{i}{2}\int d^4x\, d^4y\, J_\mu(x)\, \Delta^{\mu\nu}(x-y)\, J_\nu(y)\right]$: in position space the sources carry x and y, with $\Delta^{\mu\nu}(x-y)$ as defined below.

And of course, since $k^\mu J_\mu(k)$ is zero we can choose

$$P^{\mu\nu} = g^{\mu\nu} - \frac{k^\mu k^\nu}{k^2} \longrightarrow g^{\mu\nu}$$

~~Feynman Gauge~~ Feynman Gauge.

Lorentz Gauge

(or also known as) Landau Gauge

So here we have the free theory path integral and understand that there is some freedom in the propagator. $\Delta^{\mu\nu}(k)$

This motivates the R_ξ gauge. that

$$\Delta^{\mu\nu}(k) = \frac{1}{k^2 - i\varepsilon}\left(g^{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2}\right). \quad (1)$$

So $\xi = 1$ or 0 gives use the above gauge choices but other choices are fine, too.

And physical result shall not depend on the choice of ξ. This allows use to prove many useful aspects

(this requires adding ξ-gauge term $-\frac{1}{2\xi}$ ~~$(\partial_\mu A^\mu)^2$~~ $(\partial^\mu A_\mu)(\partial^\nu A_\nu)$

and a gauge choice is done by the gauge fixing function in the path integral)

1 Where R_ξ comes from: insert $\delta(\partial \cdot A - \omega)$ (Jacobian $\det(-\partial^2)$, A-independent) and average over ω with weight $e^{-i\int \omega^2/2\xi}$. This adds $-(\partial \cdot A)^2/2\xi$, and $k^2 g^{\mu\nu} - (1 - 1/\xi)k^\mu k^\nu$ has the inverse $\tilde{\Delta}^{\mu\nu}$. ξ multiplies only $k^\mu k^\nu$, zero on conserved currents.

Continues Symmetries.

With a lagrangian density $\mathcal{L}$, $S = \int d^4x\, \mathcal{L}(x)$

the variation would be, for d.o.f $\varphi_a(x)$

$$\frac{\delta S}{\delta \varphi_a(x)} = \frac{\delta \int d^4y\, \mathcal{L}(y)}{\delta \varphi_a(x)}$$

$$= \int d^4y \frac{\delta \mathcal{L}(y)}{\delta \varphi_a(x)}$$

$$= \int d^4y \left[\frac{\partial \mathcal{L}(y)}{\partial \varphi_b(y)} \frac{\delta \varphi_b(y)}{\delta \varphi_a(x)} + \frac{\partial \mathcal{L}(y)}{\partial(\partial_\mu \varphi_b(y))} \frac{\delta(\partial_\mu \varphi_b(y))}{\delta \varphi_a(x)} \right]$$

$$= \int d^4y \left[\frac{\partial \mathcal{L}(y)}{\partial \varphi_b(y)} \delta_{ab} \delta(x-y) + \frac{\partial \mathcal{L}(y)}{\partial(\partial_\mu \varphi_b(y))} \partial_\mu \delta_{ab} \delta(x-y) \right]$$

$$= \frac{\partial \mathcal{L}(x)}{\partial \varphi_a(x)} - \partial_\mu \frac{\partial \mathcal{L}(x)}{\partial(\partial_\mu \varphi_a(x))}$$

Sign flips next line by IBP.

Stationary & "minimal" action has

$\delta S = 0$ so the above is zero.

So

$$\frac{\partial \mathcal{L}(x)}{\partial \varphi_a(x)} \to \partial_\mu \frac{\partial \mathcal{L}(x)}{\partial(\partial_\mu \varphi_a(x))} + \frac{\delta S}{\delta \varphi_a(x)}$$

Note the difference between partial derivative ∂ and functional variation.

On the otherhand, with $\mathcal{L}(x) = \mathcal{L}(\varphi_a(x), \partial_\mu \varphi_a(x))$.

An infinitesimal transformation of field

$$\varphi_a(x) \to \varphi_a(x) + \delta\varphi_a(x)$$

$$\mathcal{L}(x) \xrightarrow[\text{goes to}]{\text{then}} \mathcal{L}(x) + \delta\mathcal{L}(x)$$

$$\delta\mathcal{L}(x) = \frac{\partial \mathcal{L}}{\partial \varphi_a(x)} \delta\varphi_a(x) + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_a(x))} \partial_\mu \delta\varphi_a(x)$$

Combining with the previous page; we get.

$$\delta\mathcal{L}(x) = \partial_\mu \left(\frac{\partial \mathcal{L}(x)}{\partial(\partial_\mu \varphi_a(x))} \delta\varphi_a(x) \right) + \frac{\delta S}{\delta \varphi_a(x)} \delta\varphi_a(x)$$

combined two terms.

$$\partial_\mu (\quad) \delta\varphi_a(x)$$

$$+$$

$$(\quad) \partial_\mu \delta\varphi_a(x)$$

Defining current $\boxed{j^\mu(x) = \frac{\partial \mathcal{L}(x)}{\partial(\partial_\mu \varphi_a(x))} \delta\varphi_a(x),}$ we have.

$$\partial_\mu j^\mu(x) = \delta\mathcal{L}(x) - \frac{\delta S}{\delta\varphi_a(x)} \delta\varphi_a(x).$$

$\delta\mathcal{L}(x)$ $\qquad$ $\frac{\delta S}{\delta\varphi_a(x)} \delta\varphi_a(x)$ $\to 0$ for stationary action.

If $\delta\mathcal{L}(x) = 0$ (meaning a transformation doesn't change $\mathcal{L}(x)$ hence $\mathcal{L}(x)$ is symmetric for such a change).

We then have

$$\partial_\mu j^\mu(x) = 0.$$

Noether current is conserved., and in components

$$\frac{\partial}{\partial t} j^0(x) + \vec{\nabla} \cdot \vec{j}(x) = 0.$$

$\frac{\partial}{\partial t} j^0(x)$: change density

$\vec{\nabla} \cdot \vec{j}(x)$: current density (non-compressible fluid dynamics)

for this transformation.

Applying to E & M.

$$\mathcal{L} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} + J^\mu A_\mu .$$

$$= -\frac{1}{4} (\partial^\mu A^\nu - \partial^\nu A^\mu)(\partial_\mu A_\nu - \partial_\nu A_\mu) + J^\mu A_\mu .$$

$$\frac{\partial \mathcal{L}}{\partial A_\mu} - \partial_\nu \frac{\partial \mathcal{L}}{\partial(\partial_\nu A_\mu)} = 0 .$$

We can see.

$$J^\mu - \partial_\nu \left[-\frac{1}{2} \times 2 (\partial^\nu A^\mu - \partial^\mu A^\nu) \right] = 0 .$$

$\frac{1}{2}$ ↑ ↖ $(\ \ )^2$

$-\frac{1}{4} \times 2$

$$J^\mu + \partial^2 A^\mu - \partial_\nu \partial^\mu A^\nu = 0$$

$$J^\mu + \partial^2 A^\mu - g^{\mu\nu} \partial^2 A_\nu - \partial^\nu \partial^\mu A_\nu = 0$$

this is the Maxwell Equation

$$\partial_\nu F^{\mu\nu} = J^\mu .$$

Now add the real interactions back.

$j^\mu(x)$ is actually in EM to Noether current for the U(1) symmetry of the Dirac field.

Recall

$$\mathcal{L}_\psi = i\bar{\psi}\partial\!\!\!/\psi - m\bar{\psi}\psi$$

The Noether current

$$j^\mu = \frac{\partial \mathcal{L}_\psi(x)}{\partial(\partial_\mu \psi(x))}\delta\psi(x)$$

here $\psi(x) \to e^{-i\alpha}\psi(x)$ | $\delta\psi(x) = (-i\alpha)\psi(x)$

we can $\alpha \to e$.

$$j^\mu = e\,\bar{\psi}(x)\gamma^\mu\psi(x).$$

$e = -0.302$ in natural units.

(Heaviside-Lorentz).

fine structure constant is $\alpha = \frac{e^2}{4\pi} = \frac{1}{137}$.

(and it is running).

And we know $J^\mu = (\rho, \vec{J})$

this is exactly, e.g.,

$$\int d^3x\, \bar{\Psi}\gamma^0\Psi = \int d^3x\, \underbrace{\chi^\dagger\chi}_{\#\text{ of } e^-} - \underbrace{\xi^\dagger\xi}_{\#\text{ of } e^+}$$ [1]

So Naturally, we shall have the interaction term

$$e\bar{\Psi}\gamma^\mu\Psi A_\mu.$$

but even before renormalising the theory, we have one problem:

this term is not gauge transformation invariant [2]

$(A_\mu \to A_\mu - \partial_\mu\Gamma(x))$.

$$\mathcal{L} = \underbrace{-\frac{1}{4}F^{\mu\nu}F_{\mu\nu}}_{} + \underbrace{i\bar{\Psi}\partial\!\!\!/\Psi}_{} - \underbrace{m\bar{\Psi}\Psi}_{} + \underbrace{e\bar{\Psi}\gamma^\mu\Psi A_\mu}_{}.$$

We can extend the definition of gauge transformations to be a set of transformations, such that

$$A_\mu(x) \longrightarrow A_\mu(x) + \partial_\mu\Gamma(x)$$

Gauge transformation:

$$\Psi(x) \to e^{-ie\Gamma(x)}\Psi(x)$$

$$\bar{\Psi}(x) \to e^{+ie\Gamma(x)}\bar{\Psi}(x)$$

Then

$$i\bar{\Psi}\partial\!\!\!/\Psi \longrightarrow i\bar{\Psi}e^{-ie\Gamma(x)}\gamma^\mu\partial_\mu e^{-ie\Gamma(x)}\Psi(x)$$

$$= i\bar{\Psi}\partial\!\!\!/\Psi + e\bar{\Psi}\gamma^\mu\Psi(x)\,\Gamma(x)$$ [3]

cancelling

[1] The braces hold only for the total: normal-ordered, $\int d^3x\,\bar{\Psi}\gamma^0\Psi = N(e^-) - N(e^+)$. χ annihilates negative-helicity electrons and creates positive-helicity positrons ($m \to 0$), so $\int \chi^\dagger\chi = N(e_L^-) - N(e_R^+)$; for $m \neq 0$ it is not even conserved.

[2] Why, if LecNote21 p. 3 found $J \cdot A$ invariant for $\partial_\mu J^\mu = 0$? Here $\partial_\mu J^\mu = ie(\bar{\Psi}E - \bar{E}\Psi)$, $E = (-i\,\partial\!\!\!/ - e\gamma^\mu A_\mu + m)\Psi$: zero only for solutions. The action and the path integral need every Ψ, so Ψ transforms too.

[3] Should read $+e\bar{\Psi}\gamma^\mu\Psi\,\partial_\mu\Gamma(x)$: $ie^{ie\Gamma}\partial_\mu(e^{-ie\Gamma}\Psi) = (i\partial_\mu + e\,\partial_\mu\Gamma)\Psi$. It cancels the $-e\bar{\Psi}\gamma^\mu\Psi\,\partial_\mu\Gamma$ of p. 12's first line. Above, $A_\mu \to A_\mu - \partial_\mu\Gamma$ (the overwritten sign is $-$).

$e\bar{\Psi}(x)\gamma^\mu\Psi(x)A_\mu \longrightarrow e\bar{\Psi}(x)\gamma^\mu\Psi(x)A_\mu - e\bar{\Psi}(x)\gamma^\mu\Psi(x)\partial\Gamma(x)$ (1)

leaving the $\mathcal{L}$ invariant under such gauge transformation

One can simplify the $\mathcal{L}$ by

$$\mathcal{L} = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} + i\bar{\Psi}\not{D}\Psi - m\bar{\Psi}\Psi$$

here $\not{D}$ is the covariant derivative.

(gauge)

$$D_\mu \not{\partial} = \partial_\mu + ieQA_\mu$$ (2)

Q for electron is "-1"

2 Signs: $e = -0.302$ is the electron's charge (p. 10) and Q the charge in units of the proton's, so a field of charge $q = -eQ$ has $D_\mu = \partial_\mu - iqA_\mu$ and transforms by $e^{-iq\Gamma}$. Electron: $i\bar{\Psi}\gamma^\mu D_\mu\Psi \ni e\bar{\Psi}\gamma^\mu\Psi A_\mu$, the vertex $ie\gamma^\mu$ (LecNote23 p. 1). Its j^0 is e per electron: p. 10's "$\alpha \to e$".

We then see that

$$D_\mu\Psi \to (\partial_\mu - ie(A_\mu - \partial_\mu\Gamma)) e^{-ie\Gamma(x)}\Psi(x)$$

$$= \exp[-ie\Gamma]\not{D}_\mu\Psi$$

also $D_\mu \to e^{-ie\Gamma}D_\mu e^{ie\Gamma}$

and see $F^{\mu\nu} = \frac{i}{e}[D^\mu, D^\nu]$

In reflection, we have promoted the

global $u(1)$ symmetry of Ψ

to a local (gauged) $u(1)$ symmetry.

the introduction of $A_\mu\bar{\Psi}\gamma^\mu\Psi$ is to make the kinetic term invariant.